Residual Ternary Algebra and Bridge Series C: The Residual–Resonant Frame for the S²-11DM²ET-X Model

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Abstract

We develop the residual (ternary) discrete algebra of the S²-11DM²ET-X model. We separate it formally from the resonant schedule that shares its central integer. Packaging $N_{\text{flux}} = 4880$ flux units with seed $f_{\max} = 21$ leaves $B' = \lfloor 4859/9 \rfloor = 539$ pieces in each of nine charge sectors. This 539 is a count. The resonant schedule $18 + 521 = 539$ is an iteration length, with $18 = \lfloor e^3/\ln 3 \rfloor$. The free canonical ternary map T_3 takes 4880 to 1 in 14 steps.

On the product complex $X_\times = K_9 \times P_{B'}$ we prove that the cycle rank of the product graph is $\beta_1 = 36B' - 8 = 19396$, and that the mixed cochain η is closed. On the thin residual form complex F we prove $H^0(F) \cong \mathbb{Q}$, $H^1(F) = 0$ and $H^2(F) \cong \mathbb{Q} \cdot [s_{\alpha\delta}]$. We show that F is formal and that its essential class is unique. We prove that the class $[s_{\alpha\delta}]$ is a feature of the thin truncation: in the full product cochain complex, $s_{\alpha\delta}$ is not closed and $H^2 = 0$.

We derive three further results: the static classifier D_1 has rank one; the residual count and the resonant clock cannot be identified; and the residual core cannot force the alternating group A_5 on the natural flux quintic. For the real-quadratic field $\mathbb{Q}(\sqrt{35})$ we record the 14-dimensional level-one parallel-weight-two cuspidal newspace V_{35} as a cited database result. We isolate precisely which definitions and lemmas are missing for the bridge $A1 \to \Xi \to$ RTA, for the eleven-dimensional Einstein–Hampton programme, and for the related non-local questions.

1. Introduction

The S²-11DM²ET-X model attaches two different meanings to the integer 539, and conflating them leads to false statements. This paper takes the separation as its organising principle.

- **Residual discrete algebra** studies the combinatorics of what is left over after the flux is packaged under Principle (S). Its 539 is the cardinality B' of each of nine sector cores.
- **Resonant algebra** is a design choice. It fires a local ternary step a fixed number of times, $18 + 521 = 539$, and then stops.
- A third number, about 14, is the typical depth of the free basins of the canonical ternary map T_3. That map contracts by a factor $2/3$ on average.

In short: a count is not a clock.

The paper is organised as follows.

- §2 constructs the residual product complex and the thin form complex, computes their cohomology, and proves formality and uniqueness of the essential class.
- §3 proves the separation theorem.
- §4 treats the static classifier D_1 and the static budget S_{eff}.
- §5 develops the local path through Bridge Series C and the newspace V_{35}.
- §6 states what is presently available for the non-local path.
- §7 summarises a consistency audit of the corpus.
- §8 ranks the open steps toward a residual foundation of RTA.

Throughout, a claim is either **proved**, with a complete proof, or explicitly described as **not established from the definitions and lemmas presently available**, in which case we name what is missing. Category A denotes statements of the discrete residual algebra. Category B denotes continuum and physical interpretations, which are not used in any proof.

All cochains have rational coefficients unless stated otherwise.

2. Residual ternary algebra

2.1 Packaging

Definition 2.1. Let $N_{\text{flux}} = \lfloor e^3 \cdot 3^5 \rfloor = 4880$ and $f_{\max} = \lceil 4880/243 \rceil = 21$. The residual pool is $R = N_{\text{flux}} - f_{\max}$. The residual cardinality is $B' = \lfloor R/9 \rfloor$. Each of the nine charge sectors carries a core of B' leftover pieces.

Proposition 2.2.

- (a) $R = 4859 = 9 \cdot 539 + 8$, so $B' = 539$.
- (b) $N_{\text{flux}} - 9B' = 29$.
- (c) $B' = 7^2 \cdot 11$, and $B' \equiv 3 \pmod 8$.
- (d) $B' - 1 \equiv 2 \pmod 8$.

Proof. (a) $e^3 \cdot 243 = 4880.78\ldots$ and $4880/243 = 20.08\ldots$, so $N_{\text{flux}} = 4880$ and $f_{\max} = 21$. Then $9 \cdot 539 = 4851$ and $4859 - 4851 = 8 < 9$. (b) $4880 - 4851 = 29$. (c) $49 \cdot 11 = 539$ and $539 = 8 \cdot 67 + 3$. (d) $538 = 8 \cdot 67 + 2$. □

2.2 The product complex

Let K_9 be the complete simplicial complex on the sectors $\{0, \ldots, 8\}$, and let $P_{B'}$ be the path with vertices $0, \ldots, B' - 1$ and edges $e_i = (i \to i + 1)$. Set

$$X_\times := K_9 \times P_{B'}.$$

Its cells are:

- vertices (a, i);
- horizontal edges $(a, b; i)$ and vertical edges $(a; e_i)$;
- triangles $(a, b, c; i)$ and squares $(a, b) \times e_i$;
- prisms $(a, b, c) \times e_i$, together with higher cells.

The product differential is

$$D(\varphi \otimes \psi) = d_Q\varphi \otimes \psi + (-1)^{\deg \varphi}\, \varphi \otimes d_P\psi,$$

and $D^2 = 0$ by the graded Leibniz rule.

The basic cochains are:

- $\alpha(a, b) = \mathrm{sgn}(a - b)$;
- $\omega_2 = d_Q\alpha$;
- $\mu = B' \alpha \smile \alpha$;
- the tower label $f(i) = \tau(x_i)$ of the i-th element of the ordered core;
- $\delta f = d_P f$.

Lemma 2.3. For distinct sectors, $\omega_2(a, b, c) = \alpha(a, b) + \alpha(b, c) + \alpha(c, a)$. On ordered triples $a < b < c$ this gives $\omega_2 = -1$ and $\mu = B'$, so $\mu = -B'\omega_2$.

Proof. By definition $(d_Q\alpha)(a, b, c) = \alpha(b, c) - \alpha(a, c) + \alpha(a, b)$, and α is skew. For $a < b < c$ the three terms are $-1, -1, +1$, which sum to -1. Also $\mathrm{sgn}(a-b)\,\mathrm{sgn}(b-c) = 1$. □

Theorem 2.4 (mixed closure). The cochain $\eta := \alpha \otimes \delta f - \omega_2 \otimes f$ satisfies $D\eta = 0$.

Proof. We use $d_P\delta f = 0$ (the path has no 2-cells) and $d_Q\omega_2 = 0$. Then

$$D(\alpha \otimes \delta f) = \omega_2 \otimes \delta f \qquad \text{and} \qquad D(\omega_2 \otimes f) = \omega_2 \otimes \delta f.$$

Subtracting gives $D\eta = 0$. □

Proposition 2.5. The following hold.

- $\sum_{a<b<c} \mu(a, b, c) = 84B' = 45276$.
- $\sum_{a\neq b} \sum_i \alpha(a, b)\, \delta f(i) = 0$.
- There are 72 oriented squares over each path edge.

Proof. There are $\binom{9}{3} = 84$ ordered triples, and $\mu = B'$ on each by Lemma 2.3. Since α is skew, $\sum_{a\neq b} \alpha(a, b) = 0$, and this factor multiplies the double sum. Finally, $9 \cdot 8 = 72$. □

2.3 The cycle space

Theorem 2.6. Let $G_\times = K_9^{(1)} \,\square\, P_{B'}$ be the Cartesian product of undirected 1-skeleta. Then $G_\times$ is connected, and

$$\beta_1(G_\times) = 36B' - 8.$$

At $B' = 539$ this gives $V = 4851$, $E = 24246$ and $\beta_1 = 19396$.

Proof. There are $V = 9B'$ vertices. The edges are $\binom{9}{2} = 36$ per level, plus $B' - 1$ per sector, so $E = 36B' + 9(B' - 1)$. Any vertex (a, i) is joined to $(0, i)$ and then along the path to $(0, 0)$, so the graph is connected. For a connected graph, $\beta_1 = E - V + 1 = 36B' - 8$. □

The rank of the cycle group of the 1-skeleton is $E - \mathrm{rank}\,\partial_1 = E - V + 1$. This is the same number, so the "free cycle space" of $X_\times$ has rank exactly $36B' - 8$. A direct union-find computation of $E - V + c$ agrees with the formula for $B' \in \{1, 2, 3, 10, 100, 538, 539, 540\}$. If horizontal edges are counted as ordered pairs, the analogous count is $72B' - 8$. We use the undirected convention.

2.4 The thin residual form complex

Definition 2.7. The thin residual form complex $F^\bullet$ is the $\mathbb{Q}$-complex with the following generators.

- $F^0 = \langle \mathbf{1},\, f\rangle$.
- $F^1 = \langle h_\alpha,\, v_\delta,\, h_{\alpha f}\rangle$, where $h_\alpha = \alpha \otimes 1$, $v_\delta = 1 \otimes \delta f$ and $h_{\alpha f} = \alpha \otimes f$.
- $F^2 = \langle t_\mu,\, t_\omega,\, s_{\alpha\delta},\, t_{\omega f}\rangle / (t_\mu + B' t_\omega)$, where $t_\omega = \omega_2 \otimes 1$, $t_\mu = \mu \otimes 1$, $s_{\alpha\delta} = \alpha \otimes \delta f$ and $t_{\omega f} = \omega_2 \otimes f$.
- $F^n = 0$ for $n \geq 3$.

The differential is

$$D\mathbf{1} = 0, \quad Df = v_\delta, \quad Dh_\alpha = t_\omega, \quad Dv_\delta = 0, \quad Dh_{\alpha f} = t_{\omega f} - s_{\alpha\delta}, \quad D|_{F^2} = 0.$$

These values follow from the product rule. For example, $D(\alpha \otimes f) = \omega_2 \otimes f - \alpha \otimes \delta f$. The relation $t_\mu = -B' t_\omega$ is Lemma 2.3.

Remark 2.8. F is the brutal truncation $\sigma_{\leq 2}$ of the generator model, so $H^2(F) = \operatorname{coker}(D^1 : F^1 \to F^2)$. The generator f must lie in F^0, because the product structure requires $Df = d_P f = \delta f$.

If f is omitted from F^0, then v_δ becomes a cocycle that is not a coboundary, and $H^1 \cong \mathbb{Q}$. With f omitted, exact rank computation gives $H^\bullet = (1, 1, 1)$. With f included it gives $(1, 0, 1)$. Only the latter is compatible with formality onto $\mathbb{Q}[0] \oplus \mathbb{Q}[-2]$.

Theorem 2.9 (form cohomology). Over $\mathbb{Q}$, we have $H^0(F) = \mathbb{Q}[\mathbf{1}]$, $H^1(F) = 0$, $H^2(F) = \mathbb{Q}\,[s_{\alpha\delta}]$, and $H^{\geq 3}(F) = 0$. Moreover

$$[\eta] = [t_\mu] = [t_\omega] = 0 \qquad \text{and} \qquad [t_{\omega f}] = [s_{\alpha\delta}].$$

Proof. Eliminating t_μ, the space F^2 has basis $(t_\omega, s_{\alpha\delta}, t_{\omega f})$. In the bases above, the differentials are

$$D^0 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad D^1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D^2 = 0,$$

with ranks 1, 2 and 0.

- Degree 0. $\ker D^0 = \langle \mathbf{1} \rangle$.
- Degree 1. $\ker D^1 = \langle v_\delta \rangle = \operatorname{im} D^0$, so $H^1 = 0$.
- Degree 2. $\dim H^2 = 3 - 2 = 1$.

Next we show $s_{\alpha\delta} \notin \operatorname{im} D^1 = \langle t_\omega,\, t_{\omega f} - s_{\alpha\delta}\rangle$. Suppose $s_{\alpha\delta} = x\, t_\omega + y\, (t_{\omega f} - s_{\alpha\delta})$. Comparing coefficients gives $x = 0$, $y = 0$ and $1 = -y$, which is impossible.

Finally:

- $\eta = s_{\alpha\delta} - t_{\omega f} = -Dh_{\alpha f}$;
- $t_\omega = Dh_\alpha$;
- $t_\mu = D(-B' h_\alpha)$;

- $t_{\omega f} - s_{\alpha\delta} \in \operatorname{im} D^1$, so $[t_{\omega f}] = [s_{\alpha\delta}]$. □

Corollary 2.10 (unique essential class). Every $z \in F^2$ can be written as $z = \lambda s_{\alpha\delta} + Dw$, with $\lambda \in \mathbb{Q}$ unique.

Proof. This is immediate from $\dim H^2(F) = 1$ and $[s_{\alpha\delta}] \neq 0$. □

We write $\xi_{\text{mix}} := [s_{\alpha\delta}] \in H^2(F) \cong \mathbb{Q}$ for the mixed class. It is the coupling of sector orientation α with residual tower transport δf.

Theorem 2.11 (formality). Let $H = \mathbb{Q}e_0 \oplus \mathbb{Q}e_2$ with zero differential. The map $\iota : H \to F$ given by $e_0 \mapsto \mathbf{1}$ and $e_2 \mapsto s_{\alpha\delta}$ is a quasi-isomorphism. Hence $F \simeq \mathbb{Q}[0] \oplus \mathbb{Q}[-2]$ in $D(\mathbb{Q})$, and all Massey products in $H(F)$ vanish.

Proof. ι is a chain map, because both images are cocycles. By Theorem 2.9, $H(\iota)$ is an isomorphism in every degree.

An explicit contraction of the complement is given by the isomorphisms $\langle f \rangle \cong \langle v_\delta \rangle$ and $\langle h_\alpha, h_{\alpha f} \rangle \cong \langle t_\omega,\, t_{\omega f} - s_{\alpha\delta} \rangle$.

For the Massey products: a triple product of degree-2 classes would lie in degree 5, where $H^5(F) = 0$, and products with $[\mathbf{1}]$ are trivial. □

2.5 The full product complex

Proposition 2.12. Let $\overline{X}_\times$ be the full product $\Delta^8 \times P_{B'}$, with all simplices of the 8-simplex. Then:

- (a) $\tilde{H}^\bullet(\overline{X}_\times; \mathbb{Z}) = 0$.
- (b) On each prism, $Ds_{\alpha\delta}\big((a,b,c) \times e_i\big) = \omega_2(a,b,c)\, \delta f(i)$. Hence $s_{\alpha\delta}$ is a cocycle if and only if $\delta f \equiv 0$.
- (c) For the residual tower labels, $\delta f \in \{0,1\}$ and δf is nonzero on 241 of the 538 path edges. Hence $s_{\alpha\delta}$ fails to be closed on $84 \cdot 241 = 20244$ prisms.
- (d) η is closed and exact: $\eta = -D(\alpha \otimes f)$.

Proof. (a) The product of a simplex and an interval is contractible. (b) This is the first identity in the proof of Theorem 2.4, evaluated on the prism. (c) This is a direct evaluation of f on the ordered core: there are 243 towers, with loads 21 (twenty towers) and 20 (the rest), and the core is taken as every ninth element of the residual pool. (d) This is the product rule. □

Thus ξ_{mix} is a class of the thin truncation F, namely the generator of $\operatorname{coker} D^1$. It is not a class of the full product cochain complex. Retaining the degree-3 generator $\omega_2 \otimes \delta f$ in the generator model gives $H^2 = 0$, because then $\ker D^2 = \langle t_\omega,\, t_{\omega f} - s_{\alpha\delta} \rangle = \operatorname{im} D^1$.

No untruncated complex in which ξ_{mix} is a nonzero cohomology class is established from the definitions presently available.

3. Resonant algebra and separation

3.1 The free map

The canonical ternary map is defined by

$$T_3(n) = \begin{cases} n/3, & n \equiv 0 \pmod 3, \\ \lfloor (4n+2)/3 \rfloor, & n \equiv 1 \pmod 3, \\ \lfloor (2n+1)/3 \rfloor, & n \equiv 2 \pmod 3, \end{cases}$$

with $T_3(0) = 0$.

Proposition 3.1.

- (a) The orbit $4880 \to 3253 \to 4338 \to 1446 \to 482 \to 321 \to 107 \to 71 \to 47 \to 31 \to 42 \to 14 \to 9 \to 3 \to 1$ has 14 steps.
- (b) 10^{18} reaches 1 in 92 steps, and 1616033987 reaches 1 in 37 steps.
- (c) On $[1, 50000]$ the maximal orbit length is 55, attained exactly at 36043 and 36592.
- (d) $\{1, 2\}$ is a 2-cycle.

Proof. (a)–(c) are exact iteration. (d) $T_3(1) = 2$ and $T_3(2) = 1$. □

The branch factors $1/3$, $4/3$ and $2/3$ have geometric mean $2/3$. This is the heuristic behind the average contraction factor $2/3$, and the Lyapunov exponent $\ln(2/3) < 0$. A measure-theoretic proof that the typical depth is about 14 is not established from the lemmas presently available. That figure is empirical.

3.2 The resonant schedule

Definition 3.2. The window is $W = \lfloor e^3 / \ln 3 \rfloor$ and the body is $b = 521$. The resonant schedule fires the local ternary step exactly $W + b$ times and then stops. The counter is the stopping rule.

Proposition 3.3. $W = 18$, so $W + b = 539$.

Proof. $e^3 / \ln 3 = 18.2826\ldots$ □

Theorem 3.4 (separation). Define three functions:

- $\mathcal{R}(N, f, k) = \lfloor (N - f)/k \rfloor$;
- $\mathcal{S}(W, b) = W + b$;
- $\mathcal{L}(n) = \min\{t : T_3^t(n) \in \{0, 1\}\}$.

Then:

- (a) $\mathcal{R}(4880, 21, 9) = \mathcal{S}(18, 521) = 539$, while $\mathcal{L}(4880) = 14$.
- (b) Neither of $\mathcal{R}$ and $\mathcal{S}$ factors through the other on the data that define them.
- (c) Every object of §2 (the spaces F and $H^\bullet(F)$, and the classifier D_1 of §4) is a finite-dimensional $\mathbb{Q}$-vector space or a linear map. None of them carries an iteration parameter.

Consequently, 539 is not the free stopping time of N_{flux}, and no result of §2 implies or uses a number of iterations.

Proof. (a) follows from Propositions 2.2, 3.3 and 3.1.

(b) Compare $\mathcal{R}(4880, 21, 9) = 539$ with $\mathcal{S}(18, 522) = 540$, and $\mathcal{S}(18, 521) = 539$ with $\mathcal{R}(4889, 21, 9) = 540$. In each pair, one function keeps the value 539 while the other changes it, so neither value determines the other.

(c) Inspect Definition 2.7 and §4. □

Corollary 3.5. A finite schedule determines no cell complex, covering or local system. Any geometry attached to the schedule, to $540 = 539 + 1$, or to a phase law $2\cos\theta$ is a Category B interpretation.

Proof. A counter carries no incidence data. The phase angle is continuum-valued and is not determined by any residual datum. □

The only geometry forced by the residual core is the contractibility of Proposition 2.12(a).

4. The static classifier D_1 and the budget S_{eff}

Definition 4.1 (D_1). D_1 is a static invariant classifier on the nine charge sectors of $X_\times$, over $\mathbb{Q}$. It is independent of time, schedule and metric, so the normalisation scale is $\Lambda_0 = \emptyset$. Its invariant subspace is the span of $\xi_{\text{mix}} = [s_{\alpha\delta}]$ in $H^2(F)$. It shares no structure with the resonant counter, in the precise sense of (ND) below.

(ND) Non-identification. The definition of D_1 uses only the cells of $X_\times$ and the cochains $\alpha, \omega_2, \mu, f, \delta f$. It takes no argument from, and no value in, the counter set $\{1, \dots, W + b\}$. No map of chain complexes, graded vector spaces or sets from $C^\bullet(X_\times)$, F or $H^\bullet(F)$ to that counter set is part of the definition. The equality $B' = W + b$ is an equality of values, not a structure map.

Proposition 4.2. D_1 has rank exactly one. In particular its rank is independent of the cycle rank $\beta_1(G_\times) = 19396$.

Proof. By Theorem 2.9, $\xi_{\text{mix}} \neq 0$ and $\dim H^2(F) = 1$. The cycle rank belongs to the graph layer, a different complex (Theorem 2.6). □

Proposition 4.3. Property (ND) holds for D_1 as defined.

Proof. This is Theorem 3.4(c). □

Layer 2, the component that would implement D_1, has not yet been constructed from this definition. No result here depends on it.

Definition 4.4 (S_{eff}). S_{eff} treats $T_3 = 539$ as a static cell capacity, or hash budget. It is not a convergence rate or a stopping time. The assignment

$$\mathcal{M} : S_{\text{eff}} \to \text{Residual}_{539} \times \text{Resonant}_{539}$$

has independent projections: the residual projection depends only on $(N_{\text{flux}}, f_{\text{max}}, 9)$, and the resonant projection only on (W, b). The internal stability bound is the free attractor depth of §3.1. The 539-step schedule is an external, constant-time override. A dual-rail ternary representation of local remainders belongs to the hardware layer and is not part of the algebraic definition.

Proposition 4.5. No global congruence $\Omega \equiv 0 \pmod{539}$ holds as a cancellation criterion.

Proof. In $\mathbb{Z}[\mathbb{Z}/3]$, Proposition 5.3 below exhibits a cancelling configuration whose determinants are 2 and 36, which are nonzero. The quantity Ω attached to it is 2, and $2 \not\equiv 0 \pmod{539}$. A winding functional W that would make Ω a criterion is not defined in the lemmas presently available. □

5. The local path: Bridge Series C

5.1 From packaging to Bridge Series C

The packaging integers factor through 7, 11 and 3^5:

- $B' = 7^2 \cdot 11$;
- $18 = 7 + 11$;
- $68 = 3 \cdot 11 + 5 \cdot 7$;
- $67 = 7 \cdot 11 - 5 \cdot 2$;
- $243 = 3^5$.

These factorisations single out the residual parameters $s \in \{7, 8, 11\}$. For such s, let

$$g_s(q) = \sum_{n \geq 0} \frac{q^{n^2}}{(-q^s; q^s)_n}.$$

For $s = 1$ this has the shape of Ramanujan's fifth-order mock theta function f_0. The orthant counts are $P_s = (-q^s; q^s)_\infty \, g_s$. Let H_s be the cone-weighted series on the associated indefinite lattice of signature $(1, 1)$. The coordinate orthant is not a cone chamber of that lattice, and **Bridge Series C** measures the mismatch:

$$C := P_{\text{orth}} - H_s.$$

For $s = 7$ the quadratic form is $Q(n, k) = n^2 + 7nk + \frac{7}{2}k(k+1)$. The series splits as $C_7 = C_{\text{axis}} + C_{\text{third}}$, with coefficients in $\{0, 1, \frac{3}{2}\}$ and support on the residues $0, 1, 2, 4 \bmod 7$. The residual shadow Ξ is a weight-$3/2$ series. Its Eichler integral, together with the axis period, is the candidate residual period of C. The published theorems on g_s and C [R1, R2] are used with the hypotheses stated there.

5.2 The field $\mathbb{Q}(\sqrt{35})$ and the newspace V_{35}

Proposition 5.1. Let $K = \mathbb{Q}(\sqrt{35})$, of discriminant 140, with fundamental unit $\varepsilon = 6 + \sqrt{35}$.

- (a) $N(\varepsilon) = 1$ and $\operatorname{Tr} \varepsilon = 12$.
- (b) The regulator is $\ln \varepsilon = 2.477888\ldots$
- (c) With $h_K = 2$, $\operatorname{Res}_{s=1} \zeta_K = 2h_K \ln \varepsilon / \sqrt{140} = 0.837679\ldots$
- (d) The odd residues r modulo 140 with $(140/r) = -1$ are 3, 11, 27, 37, 39, 41, 47, 51, 53, 57, 61, 69, 71, 79, 83, 87, 89, 93, 99, 101, 103, 113, 129 and 137. The inert primes below 100 are $3, 11, 37, 41, 47, 53, 61, 71, 79, 83, 89$.

Proof. (a) $(6+\sqrt{35})(6-\sqrt{35})=1$. (b)–(d) are direct evaluation. In (d), the Kronecker symbol at odd moduli equals the Jacobi symbol. □

The narrow class number is $h^+=4$, with $\mathrm{Cl}^+(K)\cong C_2\times C_2$. Representatives of the four classes are (1), $\mathfrak{p}_2=(2,1+\sqrt{35})$, $\mathfrak{p}_{19}$ and $\mathfrak{p}_5=(5,\sqrt{35})$.

Cited Result 5.2 [LMFDB]. The cuspidal Hilbert newspace of level (1) and parallel weight $[2,2]$ over K, summed over all four narrow components, has dimension 14. It decomposes into four newforms:

Newform	Dimension	CM	Base change
a	2	CM	yes
b	2	CM	no
c	2	CM	yes
d	8	non-CM	yes

We denote this space by V_{35}. The subscript names the field, not the dimension. The F1–F3 results below are combinatorial and scale-free, so no normalisation enters them ($\Lambda_0=\emptyset$). If a Hecke deformation requires a scale, the canonical invariant of K is $\operatorname{Res}\zeta_K=L(1,\chi_{140})$.

5.3 Cancellation lemmas (F1–F3)

The following are established results, restated with their exact hypotheses [R3].

- **F1.** The RPP cancellation problem is weakly NP-complete when the modulus M is part of the input. The result holds on the slice $M=2^k+1>\Sigma$, with $C=\{0,1\}$ and unit coefficients, and admits an $O(nM)$ dynamic programme. It fails for fixed odd $M\le\Sigma$: take $M=3$, $w=\{1,2\}$ and target 0.
- **F2.** A nonzero determinant does not prevent signed cancellation.
- **F3.** $|W|\le N\le\prod_i\|h_i\|_1=\operatorname{spr}\rho(|T|)$.

Proposition 5.3 (F2 witness). In $\mathbb{Z}[\mathbb{Z}/3]$ let $h_1=[0]+[1]$ and $h_2=-[0]+[1]+3[2]$. Then $\det\rho(h_1)=2$ and $\det\rho(h_2)=36$. Yet the signed count of paths to residue 1 is $W=0$, while the number of such paths is $N=2$.

Proof. Write $\omega=e^{2\pi i/3}$, so that $\det\rho(h)=\prod_k\hat{h}(\omega^k)$.

- For h_1: $\hat{h}_1(1)=2$ and $|1+\omega|^2=1$, so $\det\rho(h_1)=2$.
- For h_2: $\hat{h}_2(1)=3$, and $-1+\omega+3\omega^2=-3-\sqrt{3}\,i$, which has modulus squared 12. So $\det\rho(h_2)=3\cdot 12=36$.

There are two paths to residue 1: $(0,1)$, with weight $+1$, and $(1,0)$, with weight -1. They cancel. □

5.4 What the bridge requires

The bridge $A1\to\Xi\to$ RTA would assert that the leading profile of C lies outside V_{35}. This is **not established from the definitions presently available**. Four ingredients are missing:

- (i) a definition of C, or of its leading component, as an object in a category containing V_{35};
- (ii) the explicit polar (Mellin–Barnes residue) system and its positivity bounds;
- (iii) a stated relation between V_{35} and the 42-dimensional monomial space V_{42} used in the written premises;
- (iv) the lemma that the residual cocycle space has dimension one.

Given (i)–(iv), the statement $C \notin V_{35}$ would become a finite linear-algebra check against the four newforms of Cited Result 5.2.

The residual law for C is likewise not established. This is the modular completion whose period matches the Fricke samples at $1/7$, $3/7$, $5/7$ and 1 under a weight-$1/2$ slash factor. The quantum modularity of g_s and of C is open as well.

The operator Π_{fast} is not established either: a domain and a single definition are missing. What is known is a general lemma: an idempotent has spectrum in $\{0, 1\}$, and Id minus a time-average is idempotent. On the other hand, $P_0 \circ \mathrm{Ad}_g$ is not idempotent in general. For example, on $M_2(\mathbb{C})$ with $g = \mathrm{diag}(1, -1)$, the eigenvalues are $\{1, 0, -1, -1\}$.

Three analytic facts constrain any estimate on this path.

- Holomorphic Hilbert newforms of parallel weight 2 satisfy the Ramanujan–Petersson bound ($\theta = 0$) [Bl]. The exponent $7/64$ of Kim–Sarnak, which concerns Maass forms, plays no role here.
- The Burgess exponent is $3/16$ [Bu].
- The elliptic curve $Y^2 = X^3 - 35X$ has conductor $78400 = 2^6 \cdot 5^2 \cdot 7^2$. It is a different object from V_{35}.

The inclusion BQP $\subseteq C_{\text{HQH}} \subseteq$ NP is not established. The stronger form, with "NP-complete" in place of NP, is false, since $\emptyset \in$ BQP is not NP-complete. Relativised separations are known [RT].

6. The non-local path

6.1 The Einstein–Hampton equation

The eleven-dimensional equation is

$$G_{MN} = 8\pi G_{11}\, T^{(EH)}_{MN}, \qquad T^{(EH)} = T(G_4) + T(M2) + T(m) + T(-) + T(\text{PBH}),$$

with no Λ_{11}. It is posed on the warped metric

$$ds^2 = e^{-2A(y)} \eta_{\mu\nu} dx^\mu dx^\nu + dy^2 + e^{2B} \hat{g}_{ab} dz^a dz^b,$$

where B is constant and $\hat{R}_{ab} = \kappa_{\text{int}} \hat{g}_{ab}$ [EH]. For this ansatz the curvature layers L1–L7 give the block equations

$$6k^2 - 3K = -\Phi/96, \qquad 4k^2 + K = \Phi/144.$$

Proposition 6.1. These two equations hold simultaneously if and only if $k^2 = \Phi/1728$ and $K = 8k^2 = \Phi/216$.

Proof. The system is linear in (k^2, K) with determinant $6 \cdot 1 - (-3) \cdot 4 = 18 \neq 0$, so its solution is unique. Multiplying the second equation by 3 and adding the first gives $18k^2 = -\Phi/96 + \Phi/48 = \Phi/96$, so $k^2 = \Phi/1728$. Then $K = \Phi/144 - 4k^2 = (12-4)\Phi/1728 = \Phi/216 = 8k^2$. □

With $N = 4880$ the flux match requires $N = 4870.45$, a deficit of 9.55 units. The following are **not established from the definitions presently available**:

- the independence of the layers;
- the field content and action beyond the metric (in particular C_3 and G_4);
- whether the halo geometry is a solution of the equations or a fitted profile;
- the reduction formula to a four-dimensional observable;
- the origin of a four-dimensional cosmological constant.

The next decisive computation is the comparison of the four-form weights $T_{\mu\nu}/g = -G^2/96$ and $T_{ab}/g = -G^2/288$.

6.2 Reverse-parity PT and non-local quantum mechanics

The reverse-parity table is specified by the test sheet RR-NL-SHEET-01. A pass criterion requires a written join between the table and an observable, and no such join is yet available. The following are not established from the definitions presently available:

- the action of P and T on M_4 versus X_7;
- whether the PT Hamiltonian is of Bender type, and if so its metric operator;
- the meaning of non-locality (a kernel, bulk entanglement, or a guidance field);
- a no-signalling theorem for the reduced four-dimensional theory;
- a Bell/CHSH discriminator from standard quantum mechanics.

No map from V_{35} to any eleven-dimensional object is defined. In particular, Res ζ_K is an arithmetic scale and not a gravitational cutoff, and the primes 2, 5 and 7 of the conductor 78400 are local arithmetic.

7. Consistency audit of the corpus

A keyword inventory over $24{,}009$ text files of the project corpus checked eighteen classes of potential inconsistency with the definitions used here:

1. superseded normalisations Λ_0 (32 files);
2. earlier definitions of V_{35} (5);
3. class number without a narrow/wide marker (60);
4. the Burgess value 0.2000 in place of $3/16$ (17);
5. the conductor 560 in place of 78400 (11);
6. a Petersson decomposition across inequivalent spaces (6);
7. incomplete inert-prime lists (9);
8. physical readings of newform dimensions (4);
9. F1–F3 cited to unpublished drafts (39);
10. Cesàro averaging without a domain (12);
11. Res ζ_K used as an unconditional constant (26);
12. Kerr status markers (4556, mostly noise);

13. assumed presence of Layer 2 (139, including neutral mentions);
14. missing hash sidecars (19);
15. conflicting copies of the same file (none);
16. over-broad complexity wording (1);
17. locked-on-open dependencies (1507, pattern too broad);
18. V_{35} tied directly to predictions (4).

The inventory is heuristic. It finds candidate files and does not adjudicate them. Every fingerprinted file checked matched its recorded hash.

Three inequivalences recur throughout the corpus:

- residue classes versus primes;
- newform dimensions versus narrow class components;
- the elliptic curve $Y^2 = X^3 - 35X$ versus the Hilbert space V_{35}.

Each of these is fixed by the definitions of §5.2.

8. Toward a residual foundation of RTA: open steps, ranked

1. A definition of C comparable with V_{35}, and the relation $V_{35} \leftrightarrow V_{42}$ (§5.4 (i)–(iii)).
2. A written link from the residual objects (T_3, O_{res}, B', the packaging identities) to the modular framework. By Theorem 3.4, this link cannot pass through identifying the count with the schedule.
3. The residual law for C, and the quantum modularity of g_s and C.
4. The lemma that the residual cocycle space has dimension one.
5. A single definition, with a domain, of a nonzero projector Π_{fast}.
6. Density and necessity statements for A_5 fibres. These require structure outside the residual core (Theorem 8.1).
7. Three-adic equidistribution for $T^\sharp$. Neither the free basins nor the residual counts supply it.
8. Complexity-theoretic containments (§5.4).
9. The construction of Layer 2 from Definition 4.1.

Theorem 8.1 (the residual core does not force A_5). Let $P(x) = x^5 - 4880x^3 + 32879x - 243$.

- (a) P is irreducible over $\mathbb{Q}$, and
$$\begin{aligned}\operatorname{disc} P &= 3189489690138633111130132536069 \\ &= 47 \cdot 1901 \cdot 788309 \cdot 4528399596133175803.\end{aligned}$$
Since this is not a square, $\mathrm{Gal}(P/\mathbb{Q}) \not\subseteq A_5$.
- (b) The objects of §2 carry no Galois action on the roots of P. Consequently no evenness condition on $\mathrm{Gal}(P/\mathbb{Q})$ follows from them.

Proof. (a) Irreducibility and the factorisation are exact computations. All exponents in the factorisation are odd, so the discriminant is not a square. For an irreducible separable quintic, $\mathrm{Gal} \subseteq A_5$ if and only if the discriminant is a square. (b) The objects of §2 are vector spaces built from sector indices and tower labels, and none of them refers to P. □

The full equality $\mathrm{Gal}(P/\mathbb{Q}) = S_5$ is recorded in [R4]. It is not needed for Theorem 8.1.

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Status: Tip Untouched. No CERT.